SC2014 : Sensors, Interfacing & Digital Control

Part 2 : Tutorial 3 – Intro to Digital Control System

Q1 : Use Initial Value Theorem to calculate the value of $x(0)$ for the following system transfer functions (Z-Transform) :

i) $X(z) = \frac{z^2+z+1}{(z+2)(z+1)}$

ii) $X(z) = \frac{z+4}{(z+2)(z+1)}$

Q2 : Use Final Value Theorem to calculate the value of $x(\infty)$ for the following system transfer functions (Z-Transform) :

i) $X(z) = \frac{z^2}{(z-1)(z-0.3)}$

ii) $X(z) = \frac{z+1}{3(z-1)(z+0.4)}$

Q3 : Consider a system described by the following difference equations :

$$y(k) + 2.1y(k-1) + 1.4y(k-2) + 0.32y(k-3) = x(k)$$

where $y(k)$, $x(k)$ are the output/input of the system.

If the input to this system is an impulse function, find the resultant output, $Y(z)$.

Q4 : A mechanical lifting system (with the following transfer function) is shown below :

$$T.F. = \frac{Y(z)}{X(z)} = \frac{1}{z(z+8)(z+1)}$$

If the input to this system is a sampled step function, find the difference equation that you will need to implement for a software simulation of this system.